

q-Sparse Averaging

What it is, why width must grow with q , and what it means for this thesis

Source: Sanford, Hsu, Telgarsky, *Representational Strengths and Limitations of Transformers*, NeurIPS 2023, [arXiv:2306.02896](https://arxiv.org/abs/2306.02896) — Definition 4, Theorems 2–4, 10, 11 and Appendix B.3.

One-line summary. In q-sparse averaging every token must fetch and average the values of q *other* tokens that it names. A single attention layer can do this exactly when its embedding width m is about q (up to $\log N$ and precision factors): enough width is **sufficient** (Theorem 2) and, with p -bit numbers, $mp \gtrsim q$ is **necessary** (Theorem 4). Fully-connected networks and RNNs need size growing with the sequence length N instead. The width needed scales with **how much each token must pull in**, not with how long the input is.

1. The task

► Definition — q-sparse average (qSA), Def. 4

Input: N tokens $x_i = (z_i; y_i; i)$, where

- $z_i \in \mathbb{R}^{d'}$ (unit ball) is the token’s **data**,
- $y_i \subseteq [N]$ with $|y_i| = q$ is a set of q positions the token **points to**,
- i is its index.

Output for token i : the average of the data at the positions it points to,

$$\text{qSA}(X)_i = \frac{1}{q} \sum_{j \in y_i} z_j.$$

A function f ε -**approximates** qSA if $\max_i \|f(X)_i - \text{qSA}(X)_i\|_2 \leq \varepsilon$ for every input X .

A toy instance ($N = 6$, $q = 2$, scalar data)

i	z_i	y_i	$\text{qSA}(X)_i$		1	2	3	4	5	6
1	3	{2, 3}	$(1 + 4)/2 = 2.5$	1		$\frac{1}{2}$	$\frac{1}{2}$			
2	1	{1, 6}	$(3 + 9)/2 = 6$	2	$\frac{1}{2}$					$\frac{1}{2}$
3	4	{4, 5}	$(1 + 5)/2 = 3$	3				$\frac{1}{2}$	$\frac{1}{2}$	
4	1	{1, 2}	$(3 + 1)/2 = 2$	4	$\frac{1}{2}$	$\frac{1}{2}$				
5	5	{3, 6}	$(4 + 9)/2 = 6.5$	5			$\frac{1}{2}$			$\frac{1}{2}$
6	9	{2, 5}	$(1 + 5)/2 = 3$	6		$\frac{1}{2}$			$\frac{1}{2}$	

Table 1: Left: the task. Right: the attention matrix that solves it — row i puts weight $1/q$ on each position in y_i and 0 elsewhere. The output is then (attention matrix) $\times$ (column of z ’s).

Why this is “the” attention task

A self-attention unit outputs, for each token, a **convex combination of value vectors**: $f(X)_i = \sum_j \text{softmax}(A)_{ij} \cdot v_j$. qSA asks for exactly such a combination — uniform weight $1/q$ on an arbitrary, input-dependent set of q positions. So qSA isolates the one thing attention is built to do (content-based routing to arbitrary positions) and asks: **how wide must the key/query space be for the routing to be precise?** Everything else (the values, the averaging) is free.

✍ Intuition

The pointer set y_i can be *any* q -subset of $[N]$ — not a local window. A convolution or RNN that mixes nearby positions has no cheap way to do this; attention can, because the query of token i can be made to “match” the keys of exactly the positions in y_i .

2. Upper bound: width $m \approx q \log N$ is enough

◆ Theorem — Fixed precision, Thm. 2

For any N , any $m \geq \Omega(d' + q \log N)$, any $\varepsilon \in (0, 1)$ and precision $p = \Omega(\log(\frac{q}{\varepsilon} \cdot \log N))$ bits, there is a single self-attention unit (with an input MLP) of embedding dimension m that ε -approximates qSA.

◆ Theorem — Infinite precision, Thm. 3

With real-valued (unbounded-precision) arithmetic, $m \geq \Omega(d' + q)$ suffices — the $\log N$ factor disappears.

How the construction works

1. **Values carry the data:** $v_j = z_j$. Then any attention row that is uniform on y_i outputs exactly $\text{qSA}(X)_i$.
2. **Keys name positions:** each position j gets a fixed key vector $k_j \in \mathbb{R}^m$ (in the paper: vertices of a polytope built from random binary vectors; Theorem 3 uses a cyclic polytope).
3. **Queries name sets:** token i 's query q_i is chosen so that $\langle q_i, k_j \rangle$ is a fixed large value if $j \in y_i$ and noticeably smaller otherwise. After a large softmax temperature, the weights collapse to $\approx 1/q$ on y_i and ≈ 0 elsewhere.

The paper builds the queries with dual certificates from compressed sensing. The following simplified picture reproduces the same scaling:

✍ Intuition

Take random keys $k_j \in \{\pm 1/\sqrt{m}\}^m$, and let the query be the sum of the keys it should hit: $q_i = \sum_{l \in y_i} k_l$. Then

$$\langle q_i, k_j \rangle = \begin{cases} 1 + \text{noise} & j \in y_i \\ \text{noise} & j \notin y_i \end{cases} \quad \text{noise} \approx \mathcal{N}(0, q/m).$$

The routing is correct if the noise is well below 1 **simultaneously for all N positions**. The largest of N Gaussians of variance q/m is about $\sqrt{2q \log N/m}$, so we need

$$m \gtrsim q \log N.$$

Each of the q keys summed into the query adds interference, and $\log N$ pays for the union over all positions. A softmax temperature $\beta \gtrsim \log N$ then pushes the weight of non-members to zero; the numbers must resolve scores of that size, consistent with the paper's precision $p = \Omega(\log((q/\varepsilon) \log N))$.

The learned solution looks like this: Figure 2 in the paper shows a self-attention unit trained on qSA ($q = 3$) converging to the predicted attention pattern, uniform on each y_i .

3. Lower bound: $mp \gtrsim q$ is necessary

◆ Theorem — Thm. 4

For sufficiently large q , any $N \geq 2q + 1$ and any $d' \geq 1$, there is a universal constant c such that if $mp \leq cq$, **no** one-layer, single-head transformer (attention plus MLPs, $T_{d,m,d',p}^{1,1}$) $\frac{1}{2q}$ -approximates qSA.

Combined with Theorem 2 (choose $p = O(\log(q \log N))$), the width is optimal up to log factors in q and doubly-log factors in N : **the necessary and sufficient width both scale linearly with q .**

The proof idea: attention is a communication channel

In one attention layer, token i influences the computation over the other tokens **only through its query vector** $Q(x_i) \in \mathbb{R}^m$. That vector is m numbers of p bits: **mp bits**. If token i 's output requires information from its own input that cannot be compressed below q bits, then $mp \geq \Omega(q)$.

The paper makes this exact with a reduction from **set disjointness** (DISJ): Alice holds $a \in \{0, 1\}^q$, Bob holds $b \in \{0, 1\}^q$, and they must decide whether some i has $a_i = b_i = 1$. Any deterministic protocol needs q bits of communication (Yao 1979).

Who	What they set or do
Alice (holds a)	The pointer set of token $2q + 1$: for each $i \in [q]$, point at one of a pair of positions, the “1” slot if $a_i = 1$ and the “0” slot if $a_i = 0$.
Bob (holds b)	The data of positions $1 \dots 2q$: the “1” slot of pair i holds $+e_1$ if $b_i = 1$, every other slot holds $-e_1$.
Then	$\text{qSA}(X)_{2q+1} = (\{i : a_i b_i = 1\} - \{i : a_i b_i = 0\}) / q \cdot e_1$, which equals $-e_1$ iff the sets are disjoint .
Protocol	Alice sends her query $Q(x_{2q+1})$ — $O(mp)$ bits. Bob knows every other token, so he computes the attention output himself and reads off the answer (threshold at $-1 + 1/q$; a $1/(2q)$ -approximation is accurate enough).

Table 2: The reduction behind Theorem 4 (Appendix B.3), re-indexed for readability. A qSA-solving layer would give an $O(mp)$ -bit DISJ protocol, so $mp \geq \Omega(q)$.

! Careful

The bound is for **one layer with one head**. It says nothing directly about deeper models, and heads enter the related bounds as a product (mpH) — the paper’s Match3 result (Theorem 7) is the multi-head version of the same argument.

4. Contrast: other architectures pay in N

Model	Size needed for qSA	Source
one self-attention unit	$m \approx q \log N$ (fixed precision); $m \approx q$ (infinite)	Thms 2, 3, 4
fully-connected network	first-layer width $\geq Nd'/2$ (for $q \leq N/2$)	Thm 10
recurrent / memory-bounded	memory $\geq (N - 1)/2$ bits, even for $q = 1$	Thm 11

Table 3: The attention cost depends on q (what each token must gather); the others depend on N (how long the input is).

The intuition for the gap: attention can route **any** position to **any** position in one step, so the cost is only the precision of the routing. An MLP must keep all Nd' inputs linearly separable in its first layer; an RNN reading left to right must remember everything it might later be asked to fetch.

5. What it means for “width”

- **Width buys addressing capacity.** m is the dimension in which keys and queries live; it sets how many positions a single query can pick out precisely. Width needed $\approx$ (number of things to fetch) $\times$ (bits to name one among N).
- **Width and precision trade.** Only the product mp appears in the lower bound — a lower-precision model needs more width, and vice versa.
- **Length is cheap.** For attention, N enters only through $\log N$ (and not at all with infinite precision).

→ **Thesis hook**

Connection to our graph task. With adjacency-row tokens, node i 's token is essentially a pointer set $y_i = \text{its neighbours}$. One layer of “average my neighbours’ states” is literally qSA with $q = \deg(i)$ — so one hop of message passing is cheap in width ($\approx \deg \cdot \log n$; our **swap** graphs have degree ≈ 2 –4).

Connectivity is different: after a few hops a node’s **reach set** has size up to n , and gathering it in one step is qSA with q growing towards n . That is the same communication logic behind Sanford et al. 2024a’s one-layer bound for connectivity ($mH = \tilde{\Omega}(n)$), and why depth (pointer doubling, $\log n$ layers) can replace width. Our Q3 measures exactly this: the critical width m^* at depth $\lceil \log_2 n \rceil$ versus n .

! Careful

Don’t read our measured widths against these formulas directly: the theorems hide constants, assume a specific precision p , and concern *representation* (does a solution exist), not *learning* (does SGD find it). Our Q3 probe shows the gap is real — a width that can represent connectivity may still fail to learn it with too little data.

6. Summary

Task	each token averages the data of q positions it names
Enough	one attention layer with $m \approx d' + q \log N$ (fixed precision)
Necessary	$mp \geq cq$ for one layer, one head (via set disjointness)
Lesson	attention width scales with information fetched per token, not with N
For us	one hop of message passing is cheap; gathering whole reach sets is not — the depth/width trade-off Q3 measures